

– TECHNICAL DATA SHEET

NN-CalNit Calcium Nitrate

Industrial-Grade Calcium Nitrate Tetrahydrate — $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$

White crystalline / granular · highly water-soluble · CAS 13477-34-4



NN-CalNit – white crystalline granular calcium nitrate, supplied in 25 kg HDPE bags / bulk

Document No.	Revision	Issue Date	Product Code
NNPL/TDS/PROD-07	1.0	1 July 2026	NN-CalNit

01 PRODUCT DESCRIPTION

NN-CalNit is an industrial-grade calcium nitrate tetrahydrate, an inorganic salt of the formula $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$. It is a white, crystalline to granular solid that is highly soluble in water and hygroscopic in nature, supplying both calcium and nitrate-nitrogen. The product is manufactured to a minimum assay of 98% with controlled impurity levels for consistent industrial performance.

NN-CalNit is used across concrete admixture manufacture (set-accelerator), wastewater odour and corrosion control, rubber/nitrile glove production, cooling-salt and heat-storage formulations, and as a calcium-nitrogen source in fertigation. It is available both as a white crystalline/granular solid ($\geq 98\%$ assay) and as a 60% concentrated liquid solution for ready dosing.

02 USES & APPLICATIONS

- **Concrete** — Set-accelerating and corrosion-inhibiting component in concrete admixtures.
- **Wastewater** — Wastewater pre-conditioning for odour (H_2S) prevention and oxygen stabilisation.
- **Rubber & latex** — Coagulant / latex-coagulant in nitrile and rubber glove manufacturing.
- **Fertigation** — Water-soluble calcium-nitrogen source for fertigation and greenhouse crops.
- **Heat storage** — Component in heat-storage and molten-salt / cold-pack formulations.

03 CHEMICAL IDENTITY

Field	Detail
Chemical name	Calcium nitrate tetrahydrate
Chemical formula	$\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$
Molecular weight	236.15 g/mol
CAS number	13477-34-4 (tetrahydrate); 10124-37-5 (anhydrous)
HS code	2834 29 90
Synonyms	Lime nitrate; Norwegian saltpeter; Nitrocalcite
Available forms	Solid (crystalline/granular, $\geq 98\%$); 60% concentrated liquid solution

04 SPECIFICATION (INDUSTRIAL GRADE)

Typical industrial-grade values consolidated from Indian manufacturer specifications. Values are indicative; a batch Certificate of Analysis accompanies each consignment.

Parameter	Specification
Assay — solid grade (as $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$)	$\geq 98.0\%$
Assay — liquid grade (as $\text{Ca}(\text{NO}_3)_2$ solution)	60 % concentrated solution
Calcium (as Ca)	$\geq 18.0\%$
Total nitrogen (as N)	$\geq 15.5\%$
Chloride (as Cl)	$\leq 0.005\%$ (50 ppm) max
Sulphate (as SO_4)	$\leq 0.03\%$ max
Iron (as Fe)	$\leq 0.002\%$ (20 ppm) max
Lead (as Pb)	≤ 20 ppm max
Water-insoluble matter	$\leq 0.01\%$ max
pH (5 % aqueous solution)	5.0 – 7.5

05 PHYSICAL PROPERTIES

Property	Value
Appearance — solid grade	White crystalline to granular solid
Appearance — liquid grade	Clear to pale liquid (60% solution)
Odour	Odourless
Solubility in water	Highly soluble (~1200 g/l at 20 °C)
Melting point (solid)	$\approx 42.7 - 44\text{ }^\circ\text{C}$ (tetrahydrate)
Specific gravity — solid	≈ 1.82
Density — 60% liquid solution	1.3 – 1.4 kg/l
Hygroscopicity	Hygroscopic — absorbs atmospheric moisture
Bulk nature	Free-flowing granules / crystals (solid form)

06 PACKAGING, STORAGE & SHELF LIFE

Item	Detail
Packaging — solid grade	25 kg / 50 kg HDPE bags with inner LDPE liner; jumbo bags / bulk to order
Packaging — liquid grade	200 kg drums (60% solution)
Storage	Store in a cool, dry, well-ventilated area; keep bags sealed to prevent moisture pickup and caking
Segregation	Keep away from combustible and organic materials, reducing agents and acids (oxidiser)
Shelf life	24 months in unopened, correctly stored packaging
Handling	Avoid moisture ingress; reseal opened bags; first-in first-out (FIFO) rotation

07 APPLICATION HANDLING

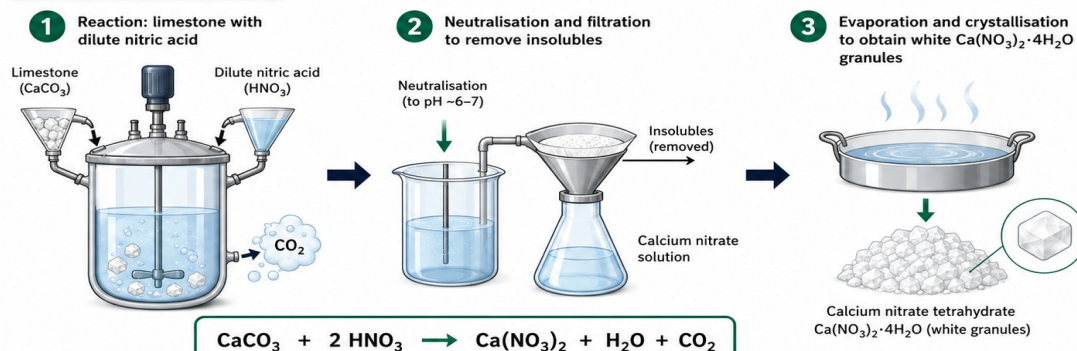


Calcium nitrate dosed as set-accelerator / wastewater conditioner at plant scale

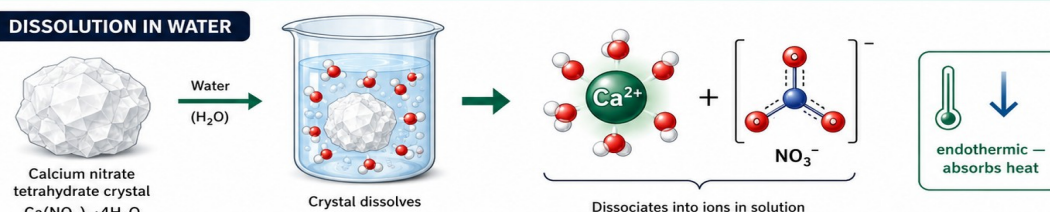
- **Dissolution** — Dissolve in water to the required concentration; the salt dissolves readily and endothermically (solution cools).
- **Dosing** — Dose per the target application (admixture formulation, wastewater loading or fertigation schedule) established by trial.
- **Equipment** — Use clean, corrosion-resistant equipment; rinse lines after use to prevent salt build-up.
- **Safety** — As an oxidiser, keep solutions and solids away from fuels, oils and organic matter.

08 MANUFACTURING PROCESS

MANUFACTURING ROUTE



DISSOLUTION IN WATER



Calcium nitrate manufacturing route and dissolution behaviour

Calcium nitrate is produced by reacting dilute nitric acid with a calcium source (limestone / calcium carbonate). The reaction — $\text{CaCO}_3 + 2 \text{HNO}_3 \rightarrow \text{Ca}(\text{NO}_3)_2 + \text{H}_2\text{O} + \text{CO}_2$ — is neutralised, filtered to remove insolubles, then evaporated and crystallised to recover the tetrahydrate, which is dried and granulated. In use, the salt dissociates fully in water into Ca^{2+} and NO_3^- ions.

09 HANDLING PRECAUTIONS

- **Oxidiser** — Oxidising agent — contact with combustible material may cause fire; store and handle away from ignition sources.
- **Irritant** — May cause mild skin/eye/respiratory irritation; wear gloves, goggles and a dust mask when handling.
- **General** — Keep away from moisture to prevent caking; keep out of reach of children and away from foodstuffs.

Refer to the NN-CalNit Safety Data Sheet for full physical, ecological, toxicological and handling information.

Disclaimer: The information and recommendations herein are given in good faith based on current knowledge. As site conditions and end-uses are beyond our control, no warranty of merchantability or fitness for a particular purpose is implied. Users must verify the product's suitability for their intended application and comply with all applicable regulations. Specification values are indicative; refer to the batch Certificate of Analysis for actual results.